Grounded Readouts of a Frozen World Action Model

Research Evolution, Literature Positioning, Experimental Protocols, and the Roadmap from RQ1a to RQ1b

WAM research project

Internal manuscript and implementation handoff

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Abstract

This document develops a research program for auditing the information available in the internal representations of a frozen DreamZero policy. It distinguishes structured temporal readout (RQ1a), grounded semantic descriptions (RQ1b), stochastic disagreement for monitoring (RQ2), and later reasoning and control (RQ3). The program has progressed beyond an initial failure-classification idea: selective extraction and validation infrastructure, a historical pilot, probe software, and an expanded collection are available. Saved independent closure records establish 150 accepted episodes; they do not establish 150 successful executions or a completed predictive study. Historical development fits provide mixed and partly negative evidence, while confirmatory evaluation on the final collection and semantic annotation remain outstanding. A targeted literature review shows substantial prior coverage of failure probes, symbolic states, latent-to-language alignment, and uncertainty monitoring. Accordingly, the proposed contribution is a controlled, temporally grounded analysis of a joint video–action policy, with explicit tests of anticipation and semantic information beyond structured outputs and language priors. We specify data boundaries, labels, baselines, grouped evaluation, annotation quality controls, staged completion criteria, and a finite resource proposal. Completed engineering, observed pilot results, planned experiments, and speculative extensions are labeled separately throughout.

Document status. This is a detailed research-design manuscript, not a report of completed final-150 experiments. It updates planning documents and preserves frozen experimental evidence. No new model training, annotation inference, data upload, rollout, or policy intervention was performed to write it. Resource envelopes are proposals for the next execution phase.

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1 Research objective and interpretation of the evidence

The central question is whether the internal representations of a frozen world action model support useful, verifiable descriptions of what is happening and what will happen next. Failure prediction is one operational consequence of this question, but it is too narrow to characterize the representation. A policy might encode present contact without anticipating a contact transition, predict eventual success through a task prior without tracking the current interaction, or generate a fluent description that contributes no information beyond its instruction. These possibilities require separate tests.

Our setting is a frozen DreamZero checkpoint evaluated on three related LIBERO spatial bowl-placement tasks. DreamZero provides the architectural motivation because it models video and action jointly[1]. The local checkpoint is identified as `dreamzero-5b-libero-step18000`; the scale and capabilities of the public flagship model must not be silently substituted for this local experimental identity. Our study concerns the representations and deployment configuration actually captured.

We distinguish four levels of evidence. *Engineering validity* establishes that the correct features, timestamps, labels and artifacts were recorded. *Decodability* establishes that a specified readout can recover a measured property on independent data. *Anticipatory utility* requires improvement over strong temporal and prior baselines before the relevant future is observed. *Causal use* requires interventions establishing how a representation or readout affects behavior. The present program targets the first three; it does not infer the fourth from predictive accuracy.

The word "interpretation" is used operationally: descriptions must be supported by observable or measured facts and must expose their temporal horizon. It is not a claim that a language decoder narrates DreamZero's private reasoning. A negative result with a small readout is also bounded: it shows a limitation of the tested representation, pooling, sample, decoder and evaluation, not the absence of all relevant information from the backbone.

1.1 Research questions

RQ1a: structured temporal readout. Which current physical relations, next-boundary relations and episode outcomes are decodable from the frozen policy, and does next-state prediction improve on persistence and class priors?
RQ1b: grounded semantic readout. Can an external readout produce faithful descriptions of the current interaction and anticipated next transition that contain information beyond the six structured probabilities and language/task priors?
RQ2: stochastic disagreement. At identical pre-decision context, does disagreement between decoded stochastic WAM samples add risk information beyond nominal risk, mean risk, action dispersion and time?
RQ3: reasoning and control. Can a unified reasoned output support planning, action revision or intervention, with causal evidence of benefit? This remains deferred.

2 How the project evolved

The following chronology reconstructs the research trajectory from dated internal plans and saved evidence. The initial conceptual stage is not assigned an invented start date. Subsequent dates refer to documentation or operational milestones, not publication claims.

Stage	Established work or framing	Scientific significance and boundary
Initial concept	Failure prediction from policy internals; stochastic uncertainty trigger; later language and planning.	A useful starting application, already substantially covered by related work.
28 September–5 October: Stage0.5	Selective extraction, pooling/parity work, validation and a historical package of 12 accepted episodes from 13 attempts.	Makes the representation study possible; engineering acceptance does not prove anticipation.
6 October: expanded RQ1	Four target families, six outputs, grouped evaluation, linear/MLP comparisons and a proposed expansion to 120 episodes.	Separates current understanding, short-horizon anticipation and outcome prediction.
6–12 October: Stage1 foundations	Data, preprocessing, probe, metric, baseline and training/evaluation modules; a combined-view smoke and three development fits.	The project already has functioning probe machinery and limited empirical evidence.
10–27 October: collection expansion	Batched collection, validation/retention work, extension toward 150 and bounded recovery procedures; final groups of 10 and 8.	Operational progress and preserved failures, not additional predictive discoveries.
28 October: final collection closure	Saved independent checks establish 150 accepted episodes and authentic original-group composites at checkpoint 787.	Completes the collection milestone; final scientific analysis remains separate.
28 October: research revision	RQ1a structured audit plus RQ1b semantic descriptions, with strengthened literature positioning and staged gates.	Descriptive language moves into RQ1b; unified planning and causal control stay in RQ3.

The change is a refinement of scope rather than abandonment of earlier work. The six-output interface remains useful as an objective anchor, diagnostic instrument and baseline for language. RQ1b tests whether additional semantic content is recoverable; simply converting six probabilities into a sentence does not by itself establish that claim. The original uncertainty-trigger idea becomes a controlled RQ2 experiment, rather than being assumed to work because several noise samples disagree.

2.1 What is completed and what is not

Observed. Saved closure 786 establishes the final eight episodes and total 150; saved independent 787 records completion with all five authentic extension composites. The local checkpoint head is 787. These statements concern authenticated collection evidence. They do not assert that every episode achieved the task goal, that final evaluation supports a hypothesis, or that a new formal dataset release has been issued.

Observed. Existing software includes data adaptation, preprocessing, linear/MLP probes, masked loss, baselines, grouped metrics, training, safe checkpoint reopening, and dataset tooling. Historical reports describe targeted tests and implementation acceptance. Their presence is an asset to reuse, not a substitute for checking compatibility with the final analysis snapshot and actual execution checkout.

Planned. The final 150 analysis input binding, support accounting, controlled fit manifest and confirmatory result card still require explicit completion evidence. RQ1b needs an annotation protocol, human-checked pilot, semantic readout and independent evaluation. RQ2 requires new same-context sampling instrumentation; RQ3 remains unimplemented. Final resource reconciliation is also outstanding, although a pre-final accounting snapshot exists.

3 Related work and coverage of the proposed contribution

3.1 Review method and limitations

The review covers the four user-supplied papers, the references in the supplied generated survey, and adjacent primary sources identified by title and citation searches, as of 28 October 2026. Closest papers were examined at method/appendix level; peripheral papers are explicitly marked as abstract-screened. This is a documented targeted review, not a certified exhaustive search of every publication. Exact reproduction requires a later full method, code, license and configuration audit. No numerical result from another task or policy is treated as a directly comparable benchmark result here.

3.2 Failure readouts and world-model representations

SAFE already learns failure detectors from policy internal features, including MLP and temporal variants, and evaluates warning behavior with conformal calibration[2]. Consequently, "train an MLP on hidden states to predict failure" is a baseline formulation. Its cumulative monitoring scores should not automatically be interpreted as calibrated Bernoulli probabilities. Its stochastic-action comparisons also limit broad claims that sampling-based monitoring would be new.

FARM is close to the frozen-representation setting: it compresses world-model tokens into a compact risk representation and evaluates both seen and strictly unseen tasks[3]. The distinction between task-pooled and strict-transfer performance is directly relevant to our small set of spatial variants. Its attention pooling cannot be reproduced exactly from our already spatially averaged video arrays. A borrowed principle, such as controlling readout capacity, must be labeled separately from a faithful implementation.

Foresight uses action-conditioned world-model predictions for failure monitoring[4]. The method's frozen encoder and trained predictor should not be conflated with an entirely frozen model. Its hidden-versus-predicted representation comparisons show that the value of future features is an empirical question. Our distinction is the controlled readout of an existing joint video–action policy under explicit current/next timing, not the invention of predicting failure from a future latent.

Beyond Task Success compares representations of several WAMs and VLAs, including multi-target behavioral diagnostics[5]. Therefore, additional targets are not sufficient novelty. Its appendix probes use episode-aggregated features and behavioral targets; that estimand differs from a pre-action decision-level prediction of the actual next boundary. DreamZero is discussed in the broader model taxonomy but is not among the evaluated models listed in that comparison. This identifies a setting to study, not proof of methodological priority. Its time-reversal analysis also cautions against reading a predictive-looking feature metric as proof of temporal causality.

3.3 Semantic interpretation and language: especially close prior art

FOREWARN already aligns world-model latent predictions with language and uses the resulting information for policy steering[6]. We borrow the need for explicit latent-to-language alignment and observation-grounded evaluation. A descriptive decoder attached to a world model, and even language-mediated candidate selection, are thus not generic novelty claims for RQ1b or RQ3.

Rogers et al. connect OpenVLA outcome prediction to SAE features, localized behavioral intervals and VLM-generated feature descriptions checked on held-out clips[7]. This directly overlaps the proposed interpretability motivation. Their appendix trains the SAE using all rollouts; our strict inductive evaluation instead requires learned representations and readouts to exclude held-out groups. Our proposed unit is a temporally specified decision and semantic target, rather than a natural-language interpretation assigned to a sparse feature. Neither approach alone establishes the cause of failure.

ECoT constructs synthetic embodied reasoning using visual and robotic evidence and then trains a policy with reasoning outputs[8]. Its annotation decomposition is useful inspiration. Our teacher is an offline labeling aid and our backbone remains frozen; the presence of a VLM-generated annotation does not turn it into ground truth or a novel method by itself.

3.4 Additional coverage and practical lessons

Work / reading scope	Existing coverage	Implication here
Lu et al.[9]; abstract	Symbolic object/relation/action-state probing of OpenVLA on LIBERO spatial tasks.	Present-state decoding is prior art; similar domain is useful context.
Progress Probes P ^{2}[10]; relevant method	Success-only progress readouts from policy internals.	A normalized time proxy is not a native physical predicate.
ActProbe[11]; abstract	Failure monitoring from action-chunk temporal consistency and magnitude.	Add cheap action-space diagnostics; do not call a partial baseline a reproduction.
FIPER[12]; overview	Success-calibrated observation-OOD and action-entropy warnings.	OOD and failure differ; broad uncertainty-trigger novelty is unavailable.
Ward et al.[13]; abstract	Latent-dynamics uncertainty for bimanual failure monitoring.	RQ2 must show incremental value in its specific controlled setting.
ContactGuard[14]; abstract	Action-conditioned pre-contact consequence monitoring and abort.	Warning before action commitment is established prior work.
Mahato and Ren[15]; abstract	Retrospective activation separation under visual shift, with shared tasks/resets/seed and clean false alarms.	Do not treat a high retrospective AUROC as independent-episode validation.

Work / reading scope	Existing coverage	Implication here
Walker et al.[16]; abstract	Jacobian-centroid monitoring of JEPA planning responsiveness.	Activation decodability and functional responsiveness are distinct; saved arrays lack Jacobians.
Latent Safety Filters[17]; abstract	Latent reachability and safety-preserving intervention.	A statistical probe is not a safety certificate.
CureWM[18]; relevant experiments	Optimistic futures and weak action sensitivity in evaluated world models.	Separate calibration from discrimination; findings do not automatically transfer to DreamZero.
Vafa et al.[19]; evaluation argument	Conventional diagnostics can coexist with incoherent implicit world models.	Limit claims to tested readout/transition utility.
Kantamneni et al.[20]; abstract	SAE probing does not consistently beat dense baselines in tested LLM regimes.	Start with strong dense controls; no universal anti-SAE robotics conclusion.

The survey's reference to "SAFE-CAST" could not be resolved to a verified robotics primary source during this search. It is excluded from supporting evidence instead of being silently substituted by an unrelated same-name work. Similarly, broad recommendations in generated summaries are not accepted as findings without a primary source.

3.5 Contribution statement and claim ladder

Proposed. The contribution is a temporally grounded, controlled audit of a frozen joint video–action policy: it separates current-state reconstruction, actual next-boundary anticipation and episode outcomes; compares representations under a fixed small-readout protocol; and tests whether a semantic interface adds verifiable information beyond structured outputs and language priors. All parts must earn support experimentally.

Possible statement	Evidence required	Present status
The collection is valid	Independent data closure and membership evidence	Established for 150 episodes
The tested features encode the selected facts	Independent grouped readout performance with controls	Historical pilot only; final study pending
The readout anticipates state changes	Advantage over priors/persistence, including transitions and timing checks	Not established
Descriptions add latent semantic information	Matched bottleneck/prior controls, independent facts and held-out evaluation	Planned RQ1b
Disagreement improves monitoring	Same-context sampling and incremental risk/latency comparisons	Planned RQ2
The model reasons causally or improves control	Appropriate interventions and controlled policy outcomes	Deferred RQ3

Using DreamZero rather than a VLA is a useful experimental setting. Predicting four families

rather than one is a useful protocol choice. Neither alone is a strong novelty argument. The paper should make the combined, bounded empirical claim that the evidence supports, including a negative or inconclusive result when that is the outcome.

4 Dataset, temporal semantics and analysis population

4.1 Collection identity and grouping

The prepared 150 membership protocol identifies 150 unique reset groups across task IDs 0, 2 and 4, with 50 groups per task. These tasks are related spatial variants of placing a black bowl on a plate. The collection therefore supports conclusions about new initial conditions within this restricted family. It does not, by itself, support generalization to unseen goals, objects, embodiments or deployment domains.

The environment seed is 0 and the nominal sampler seed is 123 under the frozen protocol. A reset-content group, including its aliases, technical attempts and any future seed-family members, is the unit of data separation. A replacement attempt is not a new independent scientific episode. More decisions within an episode, more training seeds, or more stochastic predictions at the same context do not increase the independent reset count.

Population	Train	Validation	Test	Total
Each of tasks 0, 2, 4	30	10	10	50
All groups	90	30	30	150
Legacy groups	6	3	3	12
New groups	84	27	27	138
Prospective test groups	—	—	27	27

Table 3: Membership counts derived from the immutable prepared 150 protocol. The three historical test groups remain diagnostic. These counts do not assert that every target has both classes or that an analysis-ready scoped snapshot has already been created.

The original 120 proposal had 72/24/24 groups. It is historical and must not be copied into the final 150 result table. Likewise, the presence of a membership file is different from a final training receipt: the analysis must bind each accepted episode and selected feature/label reference to the appropriate scoped input artifact.

4.2 Observation, generation and execution boundaries

Let e denote an episode, t a decision index, and $c_{e,t}$ the context available before its action chunk is executed. The frozen policy produces a joint generation and internal representation $z_{e,t}$ from this context. The experiment records 30 decisions over a 480-control horizon, with 16 actually executed controls per decision under the fixed contract.

The current label is measured at the pre-action boundary. The next label is measured after the 16 controls actually executed. It is not the next latent token, the next denoising iteration or the policy's imagined video. The extraction selector called `last_executed` identifies the selected generation forward; its name does not make a future physical observation available at prediction time.

The protocol's temporal interpretation is essential. A relation might change from 0 to 1 and back to 0 inside a chunk. Its two boundary labels would both be 0; it is not a positive next-boundary transition. An "event occurs within the chunk" target would require an independently defined

dense event label and a versioned protocol, rather than relabeling the same endpoint after seeing results.

4.3 Targets and eligibility

Family	Output	Meaning and restrictions
Current physical state	`current_on`, `current_contact`	Native bowl-on-plate and bowl–plate contact predicates at the current boundary. Contact does not mean gripper grasp.
Next physical state	`next_on`, `next_contact`	The same predicates at the next executed-chunk boundary; only valid aligned endpoints enter the target.
Ever-success outcome	`success_once`	Whether success occurs over the defined episode horizon; outcome eligibility follows the frozen first-success masking contract.
Terminal outcome	`success_end`	Whether success holds at the terminal horizon. A complete observed horizon is needed to establish a negative outcome.

There are four target families and six Bernoulli outputs. They are neither mutually exclusive categories nor six independent episodes. Preserve the established masks, including exclusion of post-first-success decisions from eligible outcome prediction under the frozen contract. An unknown or invalid cell is excluded, not silently converted into a failure or absence label.

Current/future ground truth, reward, first-success time, terminal flags, reset identifiers and elapsed time are not numeric inputs to the primary latent readout. They are labels, provenance or explicitly named diagnostic baselines. A model can otherwise obtain an attractive score through information that would not be available to a deployed monitor.

4.4 Feature representations and retained information

The stored views are video and action streams at layers 14 and 29. Each video view has shape $[2, 3072]$ after spatial pooling. The two entries are latent tokens in the recorded order; their identity must not be guessed as "current frame" and "future frame." Each action view has shape $[16, 3072]$. These arrays are the measured representation available to this study.

The baseline concatenates the two video tokens, giving 6144 coordinates per layer, and averages the 16 action tokens, giving 3072 coordinates per layer. Spatial averaging cannot later be undone to support fine object localization. Action averaging removes within-chunk order. If a readout fails, these losses remain alternative explanations; attention over original spatial tokens or an order-sensitive action head would be a distinct experiment requiring suitable retained data and a new registered comparison.

The recorder's historical evidence establishes local observed camera/proprioceptive boundary capture, including main/wrist views. For RQ1b, actual selected references, camera transformations and readability must still be checked. Twenty-four online audit examples are not the entire RGB inventory, and metadata-only online tracking is not a bulk tensor or media backup.

5 RQ1a: structured readout protocol

5.1 Existing implementation and reuse strategy

The implementation already contains `stage1_data.py`, `stage1_preprocess.py`, `stage1_probe.py`, `stage1_baselines.py`, `stage1_metrics.py`, `stage1_train.py`, `stage1_evaluate.py` and `stage1_dataset.py`. Historical reports describe deterministic preprocessing, masked training, safe checkpoint serialization, scoped train/validation inputs and controlled test access.

The next task is therefore compatibility and execution on the expanded evidence, not reimplementation of an MLP from scratch. Inspect the actual native checkout and source identity before a future run. If the final 150 manifest is incompatible with the historical release adapter, introduce only the necessary reviewed source-identity adapter. Do not impersonate a new mixed-protocol snapshot as the original pilot release. Tests should target changed contracts and unresolved concerns; already accepted payload closure is not rerun for convenience.

5.2 Preprocessing, architecture and optimization

Normalization statistics are estimated on unique training decision IDs only, with the existing minimum scale of 10^{-6}. Each view is projected into 64 dimensions using a fixed Rademacher matrix, with entries $\pm 1/\sqrt{d_{\text{in}}}$ and the predefined view seeds 31415–31418. Save the transform and its identity. Do not search for a favorable projection seed using validation or test.

There are five view choices: the four individual views and their concatenation. The linear readout maps $D \to 6$. The MLP maps $D \to 32 \to 6$, with ReLU and no dropout. $D = 64$ for one projected view and $D = 256$ for the combined input. The combined MLP has 8422 trainable parameters. This keeps the first analysis interpretable as a small-readout study rather than an unconstrained model search.

The existing optimization contract uses AdamW, learning rate 10^{-3}, weight decay 10^{-3}, batch size 64, at most 200 epochs and validation patience 20. The checkpoint is selected by the family/episode-weighted validation BCE, not by test accuracy. Training seeds 11, 22 and 33 are reported individually and together. Seed variation is optimization variability, not new experimental population size.

Let $m_{e,t,h}$ indicate that head h has an eligible target. For a family f, define an eligible per-decision loss by averaging its available heads, and then average over eligible decisions within each episode. The intended weighting can be written schematically as

$$\mathcal{L} = \frac{1}{|\mathcal{F}_+|} \sum_{f \in \mathcal{F}_+} \frac{1}{|\mathcal{E}_f|} \sum_{e \in \mathcal{E}_f} \frac{1}{|\mathcal{T}_{e,f}|} \sum_{t \in \mathcal{T}_{e,f}} \frac{\sum_{h \in f} m_{e,t,h}\, \ell(\hat{p}_{e,t,h}, y_{e,t,h})}{\sum_{h \in f} m_{e,t,h}}. \tag{1}$$

Here ℓ is binary cross-entropy and all sets include only valid support. The inspected implementation requires joint eligibility for each pair of state predicates; asymmetric masks are rejected, so each eligible state pair receives equal within-family weight. Empty families are disclosed, and no invalid target is used to fill their denominator. The objective gives equal conceptual weight to the four families rather than allowing longer episodes or the two-head families to dominate automatically. The saved numeric weights and exact implementation remain part of each future run's reproducibility evidence.

5.3 Fit matrix and bounded amendments

The established matrix is five views × two architectures × three seeds, giving 30 fits. Three unprojected linear fits on the primary video-l29 view diagnose whether random projection discards

useful signal, bringing the registered matrix to 33. The historical combined smoke and three pilot fits are separate evidence and are not relabeled as final 150 fits.

The previous max 33 accounting convention must be reconciled explicitly before execution if it was cumulative rather than per snapshot. Additional controls are not hidden inside that count. A concrete next-phase candidate manifest is 33 main/control fits, three shuffled-training primary-view controls, and nine grouped cross-fitting models for the RQ1b probability bottleneck, for 45 new fits in total. This is a prospective proposal, not existing authorization or evidence that the fits ran. The cross-fitting component can be deferred if RQ1b fails its annotation gate.

5.4 Hypotheses, baselines and estimand

The primary proposed RQ1a effect is improvement in next-on probability quality over predicted-current persistence. For head h, an episode-weighted Brier score is

$$B_h = \frac{1}{|\mathcal{E}_h|} \sum_{e \in \mathcal{E}_h} \frac{\sum_t m_{e,t,h} (\hat{p}_{e,t,h} - y_{e,t,h})^2}{\sum_t m_{e,t,h}}. \tag{2}$$

The primary model is the video-l29 MLP. For each training seed s, compare its next-on prediction with its current-on probability used as a next-state persistence forecast. The registered effect is

$$\Delta_A = \frac{1}{3} \sum_{s \in \{11,22,33\}} [B^{(s)}_{\text{next,on}} - B^{(s)}_{\text{predicted persistence}}]. \tag{3}$$

Negative values favor the next-state head. This is the mean of seed-specific scores/differences, not the Brier score of averaged predictions. The null is that the selected readout does not improve this estimand under the declared independent-group evaluation.

Mandatory comparisons include a training-derived constant probability, a training-majority classifier for thresholded metrics, deployable predicted-current persistence, and native-current oracle persistence as a separately labeled diagnostic. Oracle comparisons use matched eligibility for model and baseline. A gain against persistence while losing to the class prior is insufficient evidence of a practically useful forecast. Include the named shuffled-training negative control and separately registered time/task or action-space diagnostics where available. A simplified action-consistency baseline is not a reproduction of all of ActProbe.

5.5 Evaluation beyond a headline score

Report eligible decisions, episodes, reset groups and classes per task/split/head before interpretation. AUROC and AUPRC are undefined without both classes; return an explicit undefined value with a reason rather than a convenient number. Brier and log-loss characterize probability quality, whereas ranking and threshold metrics answer different questions. Calibration bins need support counts;150 episodes do not support arbitrarily fine calibration claims.

For next-state heads, separate actual boundary transitions 00, 01, 10 and 11. Report performance and support in each stratum, especially the relatively rare changes. The current native predicate may index an evaluation stratum but must not become a hidden feature of the deployable model. This prevents a result dominated by unchanged states from being advertised as evidence of anticipation.

For outcome heads, report decision 0 and landmarks 8, 15 and 22 under the frozen eligibility rules. The surviving population may change across landmarks; show those counts. A late score near episode termination is not equivalent to an early warning. A warning measured relative to the horizon also does not identify the point where failure became physically unavoidable.

Use paired reset-group bootstrap intervals, with 2000 draws and task stratification where support permits. All decisions of a sampled group stay together and comparators share the draws. Report bootstrap draws that produce undefined statistics, singleton strata and uncertainty limitations. Six outputs, five views and several diagnostics are not six or more independent confirmatory discoveries. The single primary estimand and secondary analyses must be labeled before final test disclosure.

5.6 Joint test lock across RQ1a and RQ1b

The 27 prospective test groups are a scarce shared resource. Opening their RQ1a performance and then using those observations to design RQ1b would make a later claim of untouched semantic evaluation misleading. The default therefore freezes the RQ1a selection and the RQ1b schema, prompts, models and comparison before any prospective performance is disclosed. Development/validation reports can be completed earlier; both final evaluations follow the joint lock.

If early RQ1a test disclosure is prioritized, later RQ1b on the same groups must be explicitly exploratory, or a separately authorized independent holdout must be collected. A new filename or receipt cannot restore blindness. The three legacy test groups remain diagnostics regardless of this choice.

6 Historical pilot findings: what we actually know

The saved 12 October summary records video-l29 MLP development fits with seeds 11, 22 and 33, on historical validation data. These are actual preliminary results, not expected results for the final collection. Lower Brier is better.

Target	Mean model Brier	Train-prior Brier	Persistence Brier	Mean AUROC
Current on	0.197854	0.170988	—	0.700779
Current contact	0.203134	0.230247	—	0.768549
Next on	0.212291	0.177778	0.202086	0.636060
Next contact	0.225017	0.237191	0.210877	0.649795
Success once	0.223113	0.250000	—	0.568651
Success end	0.279517	0.222222	—	0.261905

Table 5: Historical validation-only pilot, transcribed from the saved three-seed summary. Persistence is shown only where it is a meaningful next-state comparator. No final 150 test claim or statistical significance is implied.

Next-on is worse than persistence by 0.010205 and worse than the constant prior. Next-contact is better than the prior but worse than persistence by 0.014140. Current-contact and success-once show some probability-quality improvement over their priors, while success-end is weak in this sample. These mixed findings argue for the planned controls and a larger independent evaluation. They do not justify claiming that the WAM already anticipates transitions, or that the terminal head is useful for RQ2.

The same summary records exact online scalar/config agreement but `strict_online_acceptance=false`, with an extra `wandb-metadata.json` file. That logging-boundary caveat must be retained separately from the existence and numeric contents of the completed fits. Historical implementation tests and a successful smoke establish mechanics; they do not turn development metrics into confirmatory evidence.

7 RQ1b: from structured targets to grounded descriptions

7.1 Why this extension is worth testing

The six structured outputs provide a measurable but restricted account of an interaction. Two decisions can share all six probabilities while differing in whether the arm is approaching, transporting, hesitating, or moving away. A useful semantic readout would discriminate such observable situations and describe an anticipated transition without inventing a cause or plan. This is an empirical hypothesis: the pooled representations and 150 episodes may not support it reliably.

RQ1b should therefore be a bounded extension with an early stop criterion. It should not begin with training a general language model or generating elaborate rationales for every episode. A compact schema makes ambiguity visible and enables comparison with objective state labels. Natural language is the interface; additional supported semantic content is the scientific quantity of interest.

7.2 Teacher, student and evidence tiers

The user's proposed workflow is retained with an important qualification: record on-policy episodes, use a VLM to propose interpretations, manually check them, then train a readout of WAM representations. The VLM's output is *silver supervision*, not ground truth by declaration. Native simulator predicates provide physical reference labels; independently adjudicated human annotations provide gold semantic evaluation where the observation supports a judgment.

The teacher may consume observed RGB and permitted metadata to label a target. The deployed student receives the prescribed pre-action representation, optionally the task instruction in a separately controlled variant, and no teacher video or future labels. A teacher labeling the realized future is an offline supervisor, not evidence that the student has access to that future. Store teacher suggestions and human corrections separately so that systematic teacher bias remains inspectable.

Evidence tier	Appropriate use	What it does not establish
Native measured	Existing physical predicates and aligned boundary facts, with valid masks.	All semantics, grasp, intention or cause of failure.
VLM silver	Candidate phase/relation/movement labels and short descriptions from a bounded visual window.	Truth, reliability or independence from language priors.
Human gold	Independent annotation and adjudication using a fixed rubric and allowed evidence.	Omniscience under occlusion or a causal mechanism inferred from a video.
Student prediction	A forecast/readout available before the actual next action chunk is executed.	A fact until compared with independent evidence.

7.3 Separate the present from the future

Construct two annotation views. The current view ends at the pre-decision boundary and contains only permitted current/past observations. The realized-next view includes observations

of the next executed chunk or boundary, and is used only for future-target supervision. Their provenance records the exact visibility window. Never give the current-description annotator a complete episode and then call its retrospective explanation a current-state label.

At deployment, a next-state sentence must be framed as an anticipation or probability-qualified prediction. In training, its target may be based on the actual next observation. The distinction is analogous to supervising a forecast with the later outcome; it is valid only when the student's inputs remain causally available at decision time.

The observed boundary pairs may be insufficient for dense intermediate events. If a slip or transient contact occurs between saved frames, use "unknown" unless another authorized, aligned measurement resolves it. Do not silently infer a dense video from two latent tokens or describe a generated future as an observed one. Likewise, a bowl touching the plate does not establish that the gripper holds the bowl.

7.4 Proposed annotation schema

Each record is keyed by the original reset group, episode and decision ID. Suggested semantic fields are manipulated entity, visible object relation, interaction phase, supported movement/stalling, next-boundary relation change, unknown reason, and a short factual sentence. Evidence fields include observed-frame references, camera/transform identity, native labels and masks, teacher model/prompt version, annotation visibility, human revisions and adjudication.

The exact phase vocabulary is frozen after development-only feasibility work. Candidate categories such as approach, transport, placement and stalled interaction are hypotheses about label usefulness, not asserted existing classes. Avoid "the robot intends to" or "failure occurs because" unless the target has evidence appropriate to those claims. A sentence may correctly describe a stalled approach while leaving the underlying mechanical cause unknown.

A sentence corresponding to a six-head template might say: "Bowl–plate contact is currently unlikely; the readout predicts contact at the next boundary." An additional semantic description might say: "The gripper is approaching the bowl without visible object transport." The latter is useful only if it is separately supported and recoverable from the student inputs. Neither sentence should be generated as an example result from this dataset before the experiment exists.

7.5 Pilot selection and acceptance gates

Proposed. Begin with 60 decision units from at least 12 training/validation reset groups. Include stable and changing predicates, visually ambiguous cases and outcome-diverse development examples where available. Record this enrichment: an event-balanced audit is not a representative estimate of deployment prevalence. No prospective-test outcomes are used to select the schema or teacher.

Two humans independently annotate the pilot before adjudication. If a second annotator is unavailable, the pilot can still inform development, but cannot claim independent inter-rater agreement. Record both original judgments, the adjudicated value and reasons for disagreement. The annotation UI and prompt should show only the evidence permitted by the field's current or future horizon, and should not show the student's prediction.

Proposed gates, to freeze before the pilot, are: at least 90% non-contradiction with native facts on decidable physical fields; at least 85% agreement on core semantic fields; at least 70% decidable coverage per core field; and at least five adjudicated examples for each retained evaluated class/event distinction. Unknown–unknown pairs must not inflate the agreement numerator. Report raw agreement, class-sensitive reliability, confusion counts and uncertainty rather than

accepting a dominant-class agreement statistic alone.

These are operational go/no-go thresholds, not a power calculation or guarantee of label validity. If the available pilot lacks the required class support, narrow the vocabulary or enlarge development annotation under a new recorded decision. Do not retain an unsupported field merely to meet a planned table. If an intrinsically ambiguous field repeatedly fails, remove it from the confirmatory claim and preserve the failed annotation evidence.

7.6 Expansion and independent gold evaluation

After the rubric and teacher protocol are locked, a bounded proposal is up to 600 training/validation decision units and up to 270 prospective-test units, with 10 test units per group. These are planning maxima, not observations already annotated. Adjacent decisions are correlated, so 270 test sentences are still drawn from 27 independent reset groups. The pilot's measured annotation time and semantic coverage determine whether expansion is worthwhile.

The representative evaluation sample should be selected by an outcome-independent rule fixed in advance, for example prespecified decision positions per group. A separately reported event-enriched diagnostic sample can examine rare transitions. Do not pool the latter unweighted into a population score. Human test annotations remain sealed from schema, prompt and model selection until the joint RQ1a/RQ1b lock.

The same VLM that created labels cannot be the sole evaluator of their correctness or the student's fidelity. Human adjudication and native physical facts provide independent anchors. A second VLM can be an auxiliary diagnostic only; agreement between models is not a substitute for gold evidence. External media upload is a separate execution choice, not authorized by writing this plan. A local annotation model may avoid that boundary, but still needs a memory, license and cost check.

7.7 Readout ladder and capacity controls

Use a staged model ladder:

1. **B0: deterministic structured rendering.** Convert calibrated/thresholded RQ1a outputs into a small, explicit template. This is the minimum useful interface and a clarity baseline.
2. **B1: semantic slots from latents.** Predict the pilot-locked fields using a compact supervised head and render them through constrained text. This separates semantic extraction from linguistic fluency.
3. **B2: optional text adapter.** If labels and B1 support justify it, train a small adapter to a frozen pretrained text decoder with auxiliary structured heads. Keep WAM frozen. Report parameter count, teacher forcing and inference decoding choices.

Training a language model from scratch on 150 episodes is not a credible default. Nor is a large model automatically necessary for a few semantic fields. The exact pretrained decoder, adapter width and licenses are deferred to the next focused brainstorm and pilot measurements, rather than being asserted as implemented.

B0 versus B1 alone is an inadequate novelty test because vocabulary and learning capacity differ. Train a six-probability-to-semantics comparator using the *same semantic targets*, hidden width and optimization/selection budget as the latent semantic readout. Include instruction-only/no-latent controls and, if instructions are added to the latent model, the matched instruction condition for the comparator. Report input dimension and parameter counts; do not conceal a capacity advantage behind the phrase "same architecture."

Training probabilities for the bottleneck comparator should be produced by grouped cross-fitting within train, not by a structured model evaluated on the exact rows used to fit it. A proposed

three-fold split keeps reset groups intact and estimates preprocessing within each fold. Three seeds require nine additional structured fits. Validation/test probabilities come from the selected full-training models. This additional fit cost is explicitly named in the future manifest rather than silently bypassing the historical fit limit.

7.8 Semantic endpoints and attribution tests

The primary proposed RQ1b effect is a paired improvement in macro-F1 over pilot-locked additional semantic fields, comparing latent and trained six-probability readouts on the same held-out units:

$$\Delta_B = \text{MacroF1}(g_z(z), s) - \text{MacroF1}(g_p(\hat{\mathbf{p}}), s), \tag{4}$$

where s contains only the semantic fields beyond the six native outputs, and $\hat{\mathbf{p}}$ is the structured bottleneck. Freeze the exact field set, class treatment and averaging before test. Gold-unknown fields are not invented targets; report their coverage. A model's abstention on a gold-decidable field must affect its recall/coverage, rather than disappearing from the denominator.

Co-report supported fact precision/recall, unsupported-assertion rate, temporal contradiction rate and unknown coverage. A positive Δ_B with many confident false claims is not a satisfactory descriptive interface. Text similarity can be a secondary measure of wording, but cannot establish factual correctness, anticipation or information in the latent input.

Use no-latent and instruction-only models to test language priors, matched bottleneck models to test information beyond structured outputs, and registered shuffled-context diagnostics to test whether descriptions depend on the correct episode. Grouped uncertainty and paired examples must reflect the original reset units. Select qualitative examples by a declared rule that includes errors; attractive sentences alone are not evidence.

7.9 What would count as completion?

RQ1b is complete at the chosen scope when there is a versioned rubric, a checked annotation corpus, an independent held-out evaluation, controlled model comparisons and an explicit answer about additional faithful information. The answer may be that a constrained template is the most reliable interface at this dataset size. A failed semantic pilot is a reason to reduce scope, not to launch more generation until the text looks convincing.

8 RQ2 and RQ3: preserve the research boundaries

8.1 Same-context stochastic sampling for RQ2

RQ2 concerns repeated predictions conditioned on exactly the same pre-decision context. New resets, new whole-episode seeds and new trainer seeds answer different questions. The proposed instrumentation canary uses $K = 4$ auxiliary samples; a later study can examine nested $K = 2, 4, 8$ prefixes under fixed declared seeds. No such experiment is claimed to have run.

Preserve observation history, instruction, proprioception, cache state and nominal RNG trajectory. Auxiliary branches must not advance the simulator or alter the actions selected for the nominal rollout. Same-seed reproducibility and nominal-with-branches versus nominal-without-branches parity are prerequisite engineering checks. A hash of a partial context is not proof of equality if a relevant hidden cache or random state is missing.

For a frozen decoder's terminal-success probabilities p_k, define mean failure risk $r = 1 -$

$K^{-1}\sum_k p_k$ and semantic variance

$$V_p = \frac{1}{K-1}\sum_{k=1}^{K}(p_k - \bar{p})^2. \tag{5}$$

An entropy-gap diagnostic $H(\bar{p}) - K^{-1}\sum_k H(p_k)$ may also be recorded. These are predictive-dispersion summaries. They are not automatically calibrated failure probabilities or epistemic uncertainty, because different noise samples may represent several valid actions, decoder instability or irrelevant variability.

Compare nominal risk, mean risk, disagreement alone, mean risk plus disagreement, action dispersion and time/task diagnostics. The primary question is the *incremental* value of disagreement. Outcomes belong to the nominal executed trajectory; do not label every alternative action sample with that trajectory's realized future as if all counterfactual branches had been executed. A fixed-action, varying-video-noise study would be a distinct design and requires sampler support.

The initial trigger remains in shadow mode. Thresholds and any temporal persistence rule are calibrated only on development groups. Report episode-level false alarms, recall, warning time, class support and incremental runtime. Conformal methods in related work motivate disciplined calibration, but do not confer guarantees on our study without the relevant exchangeability, calibration and deployment assumptions. A time lead to episode termination does not prove intervention remained possible.

8.2 RQ3 remains a separate causal project

A future unified output might combine state, anticipated consequence, confidence and a proposed next action. That creates additional questions: whether text changes action selection, whether a verifier chooses better candidates, whether a policy becomes safer or more successful, and whether feedback introduces instability. These require executed interventions and controlled comparisons, not merely a language score.

RQ3 may later study candidate ranking, reasoning-conditioned planning or revised control. It remains untouched in the present experimental program. RQ1b can prepare an interpretable interface, but its descriptive success does not establish useful reasoning, safety, or a causal explanation of DreamZero's action generation. FOREWARN and latent safety work already occupy important parts of this broader space; later positioning must address them directly rather than assuming RQ3 is open territory.

9 Current state versus completion requirements

Component	Established now	Missing for completion
Collection	150 independently accepted episodes; final original-group composites.	Final read-only accounting reconciliation; no new collection required merely to start RQ1a.
Analysis dataset	Frozen 150 membership and source evidence.	Confirm existing scoped analysis artifact or prepare a compatible derived manifest; validate supports without selecting on test outcomes.

Component	Established now	Missing for completion
Structured decoder	Existing linear/MLP, preprocessing, training and evaluation implementation; historical pilot fits.	Final-snapshot compatibility, prospective fit manifest, execution and registered evaluation.
RQ1a scientific claim	Small historical validation evidence, including negative next-state comparisons.	Controlled final 150 analysis; primary/secondary tables, grouped intervals and limitation statement.
RQ1b supervision	Proposed schema and historical observed-image capture capability.	Actual frame-reference audit for selected units, teacher choice, human pilot, adjudication and independent gold set.
RQ1b model	Structured output interface can be reused.	Semantic-slot head; matched probability bottleneck/prior controls; optional text adapter; held-out fidelity results.
RQ2	Defined conceptual same-context protocol and future comparison set.	Nominal-parity instrumentation, frozen useful decoder, sampled data and separate risk/cost evaluation.
RQ3	Conceptual future direction.	Reasoning/control design, intervention authorization and experiments; intentionally untouched.

An informative result does not have to be positive. RQ1a can conclude that current relations are more accessible than next-state change under the tested readout. RQ1b can conclude that extra descriptions are unreliable at the available coverage. These are stronger scientific outputs than reporting only attractive validation examples or expanding the model until a held-out score improves.

10 Work packages, dependencies and realistic timing

10.1 Execution sequence

Package	Work and tangible output	Dependency	Effort range
WP0	Updated internal spec, roadmap and this compiled research manuscript.	Existing evidence and literature.	Current documentation task
WP1	Final 150 analysis binding, scoped receipts, train/validation support card and unresolved-evidence list.	Saved acceptance/membership.	1–2 working days
WP2	Reuse/adapt existing trainer; frozen experiment manifest; one measured fit with reopened checkpoint.	WP1 and actual source identity.	1–2 days
WP3	Controlled RQ1a fits, validation result card, baseline/transition/landmark analysis.	WP2; bounded resources.	2–4 days plus queue

Package	Work and tangible output	Dependency	Effort range
WP4	Semantic schema,60-unit pilot, independent annotation/adjudication and go/no-go report.	Development references; human annotators.	3–5 days, partly parallel
WP5	Bounded annotation expansion, B0/B1 comparisons, cross-fitted bottleneck; optional B2.	Successful WP4; fixed labels/model selection.	10–20 days
WP6	Joint test lock, prospective RQ1a/RQ1b evaluation, error analysis and final manuscript tables.	Both selection processes frozen.	Included in combined range;2–4 focused days
WP7	Separate RQ2 canary and budget/design handoff.	Frozen evaluated decoder or explicit exploratory status.	Estimate after canary

With one focused researcher and timely access to a second annotator, an RQ1a training/validation report is plausible in 5–10 working days because the software and collection already exist. Under the default joint-test lock, that is not yet the final confirmatory test report. A coherent RQ1a+RQ1b package is a 4–6 calendar-week planning range. Queue delays, expired access, annotation disagreement, missing semantic coverage or code incompatibilities add uncertainty; no runtime benchmark from the final 150 training workload has been measured for this estimate.

An illustrative schedule is: week 1 for analysis inputs, compatibility, canary and initial matrix; week 2 for structured validation analysis and semantic-pilot adjudication; weeks 3–4 for bounded labels and semantic comparisons; weeks 5–6 for optional text adaptation, final lock/evaluation and manuscript integration. WP4 can overlap compute-heavy parts of WP2–3. Prospective test disclosure cannot be parallelized with adaptive RQ1b development while retaining confirmatory independence.

10.2 What greater effort can accelerate

The highest-value acceleration is sustained researcher time on one clear question, reuse of the existing trainer, ready access to a second annotator, and bounded parallel execution of independent fits after measuring one. More GPU credits can reduce a demonstrated inference bottleneck. They do not remove manual adjudication, create new independent tasks from repeated decisions, or repair a target whose truth is visually unobservable.

Keep a short daily decision log: what evidence changed, which comparison is now runnable, and what is blocking the next claim. A weekly result card should show actual completed runs and annotation coverage, not only infrastructure activity. Prefer a small completed comparison to simultaneous unfinished architectural extensions. Broad external-method reproduction, new-task collection, fine-tuning the WAM and RQ3 control are optional expansion projects, each with its own question and cost.

10.3 Ownership and handoff

The researcher owns questions, preregistered comparisons and claim interpretation. A single controller owns submission identities, active allocations and accounting, even if independent jobs run concurrently. Two annotators provide independent semantic judgments; an adjudicator

resolves disagreements using the written rubric. A reviewer checks temporal availability, split independence and claim/evidence alignment.

The next short brainstorming should settle the semantic vocabulary and minimal output format, choose the annotation personnel and feasible teacher, approve a concrete fit manifest and finite budget, and decide whether the near-term paper is a focused single-policy audit or a broader later benchmark. It should not reopen already verified collection simply to feel more certain. A model choice deferred until a memory/annotation canary is an intentional dependency, not an excuse to redesign the entire project.

11 Resource accounting and a more ambitious finite envelope

11.1 Historical consumption is not the new research budget

The latest available accounting snapshot is checkpoint 785, before the last source closure and final composite operations. It records all 80 known jobs terminal, an empty queue,67 completed jobs,12 failed jobs and one historical canceled job. This is a dated snapshot, not a fresh queue claim at the time of this manuscript.

Quantity at 785	Recorded value	Interpretation
GPU elapsed	140543 seconds	Reported terminal elapsed allocation, about 39.04 GPU-hours.
CPU elapsed	52160 seconds	Reported terminal CPU-allocation elapsed, about 14.49 allocation-hours; not core-hours.
Conservative GPU charge	144143 seconds	Includes 3600 seconds for a zero-elapsed historical canceled reservation. Not all measured spend.
Conservative CPU charge	204090 seconds	Includes RPC wall ceilings/startup/buffers; actual RPC 759 wall remains unknown.
Retained storage	7207671428 bytes	Measured pre-final retained inventory, not a final 787 storage figure.

Table 9: Existing pre-final accounting evidence. Do not add measured and conservative columns as if they were independent costs. Reservations are not measured expenditure. Final 786/787 cost reconciliation remains outstanding.

The last two collection groups have saved terminal evidence: group 10 GPU 28613265 completed in 11819 seconds and CPU 28619694 in 1666 seconds; the last 8 GPU 28623348 completed in 6414 seconds and CPU 28630934 in 2124 seconds. These allocations are already part of the relevant cumulative accounting history; they are not to be added again to the 785 totals. No unknown control duration is filled with an invented actual value.

The latest collection authority retained aggregate ceilings of 345600 GPU-seconds and 518400 CPU wall-seconds, with retention ceiling 137438953472 bytes; grouped prospective per-job limits were separately increased. Those operational limits and the collection admission cutoff are historical collection authority. They neither authorize unlimited subsequent research nor need to be overwritten to propose a new phase.

11.2 Proposed research envelope

The user explicitly wants greater effort and more useful Snellius expenditure. A finite, staged envelope supports this without equating larger reservations with progress:

Phase	Proposed ceiling	Admission condition
RQ1a structured study	24 CPU-allocation hours	One measured fit establishes allocation/time needs; actual core count recorded.
Optional RQ1a GPU execution	8 A100-hours	CPU benchmark demonstrates a reason to use GPU.
RQ1b annotation pilot	8 A100-hours	Teacher/model/input contract and local resource feasibility fixed.
RQ1b expansion/modeling	32 A100-hours	Pilot passes quality/coverage gates and expansion manifest is frozen.
Total GPU proposal	48 A100-hours	Ceiling across these phases, not forecast expenditure.
RQ2 or new collection	Separate estimate	Measured canary and independently specified question.

Table 10: Prospective discussion envelope. No allocation was submitted under this proposal. CPU and GPU budgets are not interchangeable and unused budget is not automatically transferred.

At 32 allocated CPU cores,24 allocation-hours correspond to 768 core-hours. A request for two cores may still result in a larger billable allocation, so report requested and allocated resources separately. Likewise, GPU-hours mean elapsed allocation multiplied by GPU count; memory, partition and account may affect actual charging. This document does not assert a euro or credit conversion without the user's account-specific tariff.

Allow at most two independent concurrent research jobs after the canary, available quota and unique ownership are verified. This is proposed concurrency, not permission to start a duplicate controller or repeat an unknown-outcome submission. Measure whole allocation wall time once, distinguish reservations from terminal spend, and avoid counting nested diagnostics twice. Larger models or more tasks require a revised finite manifest, not a silent relaxation of all limits.

12 Threats to validity and publication limits

Population scope. Three related spatial tasks and 150 reset groups cannot establish general-purpose manipulation understanding. The 27 prospective test groups are valuable but small; confidence intervals and unsupported classes must remain visible. Additional task diversity may be more scientifically valuable than repeated runs on existing resets, but is a separate future collection decision.

Representation loss. Spatial pooling, action-token averaging, random projection and a small decoder can conceal useful information. Conversely, a stronger decoder can exploit priors without showing that the WAM uses the information causally. Both limitations constrain interpretation.

Temporal and outcome shortcuts. Current predicates can be easy while change is rare; success can correlate with elapsed time or task identity; post-success observations can make outcome prediction trivial. The fixed masks, transition strata and named diagnostics are designed to reveal these effects rather than claim they are impossible.

Annotation validity. VLM errors, occlusion, phase ambiguity and hindsight can enter a semantic corpus. Human agreement is also imperfect and can be inflated by dominant classes or shared unknowns. Independent adjudication, visible coverage and temporal input restrictions are required, with explicit failure criteria.

Selection leakage. Reusing a test set across successive research questions can convert later claims into exploratory ones even when individual trainers never read test labels. The joint lock is a design-level control, not merely a loader permission.

Literature and novelty. The reviewed literature already covers many individual ingredients. A useful final contribution depends on the controlled combination and empirical findings, and the related-work section must be refreshed before submission. This manuscript does not assert priority, universal superiority or safety guarantees.

Operational versus scientific completion. Source closure, online logging acceptance, analysis snapshot preparation, model training and scientific validation are different milestones. In particular, the 150 accepted collection and historical software tests do not establish a successful final decoder or a final release. Preserve all these distinctions when adapting this text into a paper.

A Local evidence and provenance map

Paths below are relative to the workspace root. They are sources for this internal account, not public download links. Historical files remain unchanged except for dated navigation banners in the older design/plan entry points.

Evidence	Local source / role
Append-only project MASTER	`work/stage1-rq1-sdd/progress.md`; current acceptance chronology, operational limits and unresolved accounting.
Final extension events	`work/stage1-rq1-sdd/extension10-20261023/progress.jsonl`;786/787 saved completion events.
Frozen 150 membership	`work/stage1-rq1-sdd/extension30-protocol-prepared-20261016/protocol.json`;90/30/30, 27 prospective test groups and feature/temporal identity.
Historical numerical pilot	`work/stage1-rq1-sdd/meeting-three-seed-summary-2026-10-12.json`; validation-only three-seed table and online acceptance caveat.
Implementation reports	`work/stage1-rq1-sdd/task-2-report.md` and`work/stage1-rq1-sdd/task-4-report.md`; existing probe/trainer interfaces and historical verification.
Original design	`docs/superpowers/specs/2026-10-06-wam-rq1-rq2-stage3-design.md`; target semantics and historical 120 proposal.
Revised design	`docs/superpowers/specs/2026-10-28-rq1-research-revision.md`; current framing and builder-review amendments.
Execution roadmap	`docs/superpowers/plans/2026-10-28-rq1a-rq1b-research-roadmap.md`; gated future tasks and resources.

Evidence	Local source / role
Historical Stage0.5 handoff	`docs/2026-10-05-stage05-next-steps-for-bram-en.md`;12-episode engineering stage.

The saved final source proof is`1b521d18d88b180ae4cb65c572e4aba16237de457dde85671c649e98e49fd2ef`; the checkpoint 787 head is`cc20fe29489b3bfcf1923ad55e8472a90cc12b39567d074e2592a5fb3b5cb429`. These identifiers reference already saved validation, not a new hash/re-audit performed for the manuscript. The earlier full read-only reconciliation was saved as`/private/tmp/rq1-reconciliation785-after-unlock.json`. Final accounting remains separately unresolved.

B Result tables to populate after execution

The final paper should add three result blocks without retroactively rewriting this design: (i) RQ1a per-head supports and grouped effects, with transition-specific diagnostics; (ii) RQ1b annotation quality, coverage and independent semantic comparisons; (iii) resource and latency measurements from actual allocations. No placeholder numeric result in this section is intended.

For every result, record the input snapshot, group membership, model/checkpoint identity, selection rule, metric definition, eligible population, comparison and uncertainty method. Label validation, legacy diagnostics and prospective test distinctly. Report the actual number of attempted, completed and excluded fits, including negative controls and cross-fitting. Keep a separate operational failure appendix so that a failed infrastructure generation does not become a failed scientific episode and an accepted episode does not become a successful task by terminology alone.

The next decision is to freeze a feasible semantic pilot and finite run manifest, then execute the already implemented structured study and the bounded annotation work. Further architecture, uncertainty sampling and control should follow evidence rather than precede it.

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